



Date: 02.01.2025

Minutes of the First Board of Studies Meeting

The First Meeting of BOS, B.Tech in CSE(Data Science) was held at 2.30pm through online mode on 21.12.2024 (Saturday) using the following link:

<https://us06web.zoom.us/j/82075204135>

The following members attended the meeting:

S.No.	Name of the Member	Designation	Role
1.	Dr. Loshma Guniseti	Professor, HOD-AIML, SVEC	Chairperson
2.	Prof. L Sumalatha	Professor, Department of CSE UCEK, Kakinada	University Nominee
3.	Prof. R B V Subrahmanyam	Professor, Department of CSE, NIT- Warangal	Academic Expert
4.	Dr. K Himabindu	Dean, Student Welfare, NIT Andhra Pradesh	Academic Expert
5.	Mr. Pavan Kumar Chowdarapu	Delivery Partner / Solution Architect, TCS Hyderabad	Industry Expert
6.	Mr. Sattibabu Nimmakayala	Lead Consultant, Infosys Limited, Hyderabad	Alumni
7.	Dr. G. V. N. S. R. Ratnakara Rao	Principal	Invited Member
8.	Dr. Ch. Rambabu	Dean, SA	Invited Member
9.	Dr. D. Jaya Kumari	Professor & Head, CSE & CST	Invited Member
10.	Mrs. A. Leelavathi	Sr. Asst. Professor	Member
11.	Mr. R. L. Phani Kumar	Sr. Asst. Professor	Member
12.	Mr. Vemana Ramanarayana	Sr. Asst. Professor	Member
13.	Mr. M. Yesu Sekharam	Assistant Professor	Member



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14.	Mrs. K. Durga Saranya	Assistant Professor	Member
15.	Mrs. M. Kiranmai	Assistant Professor	Member
16.	Mr. B V V Bhargav	Assistant Professor	Member
17.	Mr. R Lakshmana Rao	Assistant Professor	Member

Item No. 1: Welcome note by the Chairperson BOS.

The HOD extended a formal welcome and introduced the members.

Item No. 2: Approved Course Structure and Syllabus for I and II Semesters of B.Tech in CSE(DS) Programme under V23 Regulation, along with the following suggested changes:

Sem	Course Code	Course Name	Suggestions	Inclusions / Modifications
I	V23CST01	Introduction to Programming	Elaborate the Introduction to Strings concept in Unit III	Updated the Introduction to Strings concept in Unit III to include: Traversing a String, String Manipulation, String Functions
			Change 3 rd Reference Book as Prescribed textbook.	Modified C Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, Cengage, 3 rd edition as Prescribed textbook.
			Update the Course Outcomes	Course Outcomes have been refined.
I	V23CSL01	Computer Programming Lab	Drop the Tutorial and Objective words in Syllabus.	Tutorial and Objective words have been removed from the Syllabus.



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II	V23CST02	Data Structures	Add Binary Trees and Tree Traversals to Unit V	Binary Trees and Tree Traversals topics have been included in Unit V
II	V23CSL02	IT Workshop	Include How Computers Work, Ron White, Tim Edward Downs, Que in the Reference Books	Included How Computers Work, Ron White, Tim Edward Downs, Que in the Reference Books
			Remove the phrase “Every student” in Syllabus	Syllabus has been refined by removing the words such as “Every student”

The Approved Course Structure and Modified Syllabus as per the above suggestions is given in **Annexure-I**

Item No. 3: Suggestions given by the experts to improve the skill set of students

- Encourage the students to achieve Certifications in trending technologies
- From Third Year onwards motivate the students to learn latest technologies
- Create awareness to students on how to use ChatGPT for asking questions effectively
- Arrange industry oriented sessions by industry experts
- Lab topics could be generalized to provide flexibility to the instructors to select relevant programs
- Conduct workshops for faculty members to improve their practical skills
- Provide awareness to the students on latest Data Science tools/ technologies like R, Python, and Statistics with probability, tools and techniques.
- Data Security, Data Visualizations related courses could be included in future

Chairperson of BOS
(Dr. Loshma Guniseti)

ANNEXURE-I

First Year Course Structure

I – Semester							
S.No.	Course Code	Category	Name of the Course	L	T	P	C
1.	V23ENT01	BS&H	Communicative English	2	0	0	2
2.	V23PHT01	BS&H	Engineering Physics	3	0	0	3
3.	V23MAT01	BS&H	Linear Algebra & Calculus	3	0	0	3
4.	V23EET01	Engineering Science	Basic Electrical and Electronics Engineering	3	0	0	3
5.	V23CST01	Engineering Science	Introduction to Programming	3	0	0	3
6.	V23ENL01	BS&H	Communicative English Lab	0	0	2	1
7.	V23PHL01	BS&H	Engineering Physics Lab	0	0	2	1
8.	V23EEL01	Engineering Science	Electrical and Electronics Engineering Workshop	0	0	3	1.5
9.	V23CSL01	Engineering Science	Computer Programming Lab	0	0	3	1.5
10.	V23SPT02	BS&H	Health and wellness, Yoga and sports	-	-	1	0.5
Total				14	0	11	19.5

L-Lecture, T-Tutorial, P-Practical, C-Credits

II – Semester							
S.No.	Course Code	Category	Name of the Course	L	T	P	C
1.	V23CHT01	BS&H	Applied Chemistry	3	0	0	3
2.	V23MAT02	BS&H	Differential Equations & Vector Calculus	3	0	0	3
3.	V23CMT01	Engineering Science	Basic Civil and Mechanical Engineering	3	0	0	3
4.	V23MET01	Engineering Science	Engineering Graphics	2	0	2	3
5.	V23CST02	Professional Core	Data Structures	3	0	0	3
6.	V23CHL01	BS&H	Engineering Chemistry Lab	0	0	2	1
7.	V23EEL01	Engineering Science	Engineering Workshop	0	0	3	1.5
8.	V23CSL02	Engineering Science	IT Workshop	0	0	2	1
9.	V23CSL03	Professional Core	Data Structures Lab	0	0	3	1.5
10.	V23SPT01	BS&H	NSS/NCC/Scouts & Guides/Community Service	-	-	1	0.5
Total				14	0	13	20.5

L-Lecture, T-Tutorial, P-Practical, C-Credits

FIRST YEAR SYLLABUS

L	T	P	C
3	0	0	3

INTRODUCTION TO PROGRAMMING **(Common to All branches of Engineering)**

Course Outcomes:

After Successful completion of the Course, the student will be able to:

C01: Describe basics of computers, the concept of algorithm and algorithmic thinking (K2)

C02: Illustrate the various Control Structures (K3)

C03: Demonstrate the concepts of Array and Strings (K3)

C04: Illustrate the concepts of Pointers & User Defined Data types (K2)

C05: Demonstrate the concepts of Functions & File Handling. (K3)

UNIT I Introduction to Programming and Problem Solving

History of Computers, Basic organization of a computer: ALU, input-output units, memory, program counter, Introduction to Programming Languages, Basics of a Computer Program- Algorithms, flowcharts (Using Dia Tool), pseudo code. Introduction to Compilation and Execution, Primitive Data Types, Variables, and Constants, Basic Input and Output, Operations, Type Conversion, and Casting.

Problem solving techniques: Algorithmic approach, characteristics of algorithm, Problem solving strategies: Top-down approach, Bottom-up approach, Time and space complexities of algorithms.

UNIT II Control Structures

Simple sequential programs Conditional Statements (if, if-else, switch), Loops (for, while, do-while) Break and Continue.

UNIT III Arrays and Strings

Arrays indexing, memory model, programs with array of integers, two dimensional arrays, Introduction to Strings: Traversing a String, String Manipulation, String Functions

UNIT IV Pointers & User Defined Data types

Pointers, dereferencing and address operators, pointer and address arithmetic, array manipulation using pointers, User-defined data types-Structures and Unions.

UNIT V Functions & File Handling

Introduction to Functions, Function Declaration and Definition, Function call Return Types and Arguments, modifying parameters inside functions using pointers, arrays as parameters. Scope and Lifetime of Variables, Basics of File Handling

Note: The syllabus is designed with C Language as the fundamental language of implementation.

Textbooks:

1. C Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE, 3rd edition

Reference Books:

1. Computing fundamentals and C Programming, Balagurusamy, E., McGraw-Hill Education, 2008.
2. Programming in C, Reema Theraja, Oxford, 2016, 2nd edition
3. "The C Programming Language", Brian W. Kernighan and Dennis M. Ritchie, Prentice-Hall, 1988
4. Schaum's Outline of Programming with C, Byron S Gottfried, McGraw-Hill Education, 1996

L	T	P	C
0	0	3	1.5

COMPUTER PROGRAMMING LAB

(Common to All branches of Engineering)

Course Outcomes:

After Successful completion of the Course, the student will be able to:

CO1: Demonstrate the steps in execution of programs written in C language (K3)

CO2: Select the right control structure for solving the problem (K3)

CO3: Develop C programs which utilize memory efficiently using programming constructs like pointers (K3)

CO4: Demonstrate the applications of arrays, functions, basic concepts of pointers in C (K3)

UNIT I

WEEK 1: Problem-solving using Computers.

Lab 1: Familiarization with programming environment

- i) Basic Linux environment and its editors like Vi, Vim & Emacs etc.
- ii) Exposure to Turbo C, gcc
- iii) Writing simple programs using printf(), scanf()

WEEK 2: Problem-solving using Algorithms and Flow charts.

Lab 1: Converting algorithms/flow charts into C Source code.

Developing the algorithms/flowcharts for the following sample programs

- i) Sum and average of 3 numbers
- ii) Conversion of Fahrenheit to Celsius and vice versa
- iii) Simple interest calculation

WEEK 3: Variable types and type conversions:

Lab 3: Simple computational problems using arithmetic expressions.

- i) Finding the square root of a given number
- ii) Finding compound interest
- iii) Area of a triangle using heron's formulae
- iv) Distance travelled by an object

UNIT II

WEEK 4: Operators and the precedence and as associativity:

Lab 4: Simple computational problems using the operator' precedence and associativity

i) Evaluate the following expressions.

- a. $A+B*C+(D*E) + F*G$
- b. $A/B*C-B+A*D/3$
- c. $A+++B---A$
- d. $J= (i++) + (++i)$

ii) Find the maximum of three numbers using conditional operator

iii) Take marks of 5 subjects in integers, and find the total, average in float

WEEK 5: Branching and logical expressions:

Lab 5: Problems involving if-then-else structures.

- i) Write a C program to find the max and min of four numbers using if-else.
- ii) Write a C program to generate electricity bill.
- iii) Find the roots of the quadratic equation.
- iv) Write a C program to simulate a calculator using switch case.
- v) Write a C program to find the given year is a leap year or not.

WEEK 6: Loops, while and for loops

Lab 6: Iterative problems e.g., the sum of series

- i) Find the factorial of given number using any loop.
- ii) Find the given number is a prime or not.
- iii) Compute sine and cos series
- iv) Checking a number palindrome
- v) Construct a pyramid of numbers.

UNIT III

WEEK 7: 1 D Arrays: searching.

Lab 7: 1D Array manipulation, linear search

- i) Find the min and max of a 1-D integer array.
- ii) Perform linear search on 1D array.
- iii) The reverse of a 1D integer array
- iv) Find 2's complement of the given binary number.
- v) Eliminate duplicate elements in an array.

WEEK 8: 2 D arrays, sorting and Strings.

Lab 8: Matrix problems, String operations, Bubble sort

- i) Addition of two matrices
- ii) Multiplication two matrices
- iii) Sort array elements using bubble sort
- iv) Concatenate two strings without built-in functions
- v) Reverse a string using built-in and without built-in string functions

UNIT IV

WEEK 9: Pointers, structures and dynamic memory allocation

Lab 9: Pointers and structures, memory dereference.

- i) Write a C program to find the sum of a 1D array using malloc()
- ii) Write a C program to find the total, average of n students using structures
- iii) Enter n students data using calloc() and display failed students list
- iv) Read student name and marks from the command line and display the student details along with the total.
- v) Write a C program to implement realloc()

WEEK 10: Bitfields, Self-Referential Structures, Linked lists

Lab 10 : Bitfields, linked lists

Read and print a date using dd/mm/yyyy format using bit-fields and differentiate the same without using bit- fields

- i) Create and display a singly linked list using self-referential structure.
- ii) Demonstrate the differences between structures and unions using a C program.
- iii) Write a C program to shift/rotate using bitfields.
- iv) Write a C program to copy one structure variable to another structure of the same type.

UNIT V

WEEK 11: Functions, call by value, scope and extent,

Lab 11: Simple functions using call by value, solving differential equations using Eulers theorem.

- i) Write a C function to calculate NCR value.
- ii) Write a C function to find the length of a string.
- iii) Write a C function to transpose of a matrix.
- iv) Write a C function to demonstrate numerical integration of differential equations using Euler's method

WEEK 12: Recursion, the structure of recursive calls

Lab 12: Recursive functions

- i) Write a recursive function to generate Fibonacci series.
- ii) Write a recursive function to find the lcm of two numbers.
- iii) Write a recursive function to find the factorial of a number.
- iv) Write a C Program to implement Ackermann function using recursion.
- v) Write a recursive function to find the sum of series.

WEEK 13: Call by reference, dangling pointers

Lab 13: Simple functions using Call by reference, Dangling pointers.

- i) Write a C program to swap two numbers using call by reference.
- ii) Demonstrate Dangling pointer problem using a C program.
- iii) Write a C program to copy one string into another using pointer.
- iv) Write a C program to find no of lowercase, uppercase, digits and other characters using pointers.

WEEK14: File handling

Lab 14: File operations

- i) Write a C program to write and read text into a file.
- ii) Write a C program to write and read text into a binary file using `fread()` and `fwrite()`
- iii) Copy the contents of one file to another file.
- iv) Write a C program to merge two files into the third file using command-line arguments.
- v) Find no. of lines, words and characters in a file
- vi) Write a C program to print last n characters of a given file.

Textbooks:

1. C Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE, 3rd edition

Reference Books:

1. Computing fundamentals and C Programming, Balagurusamy, E., McGraw-Hill Education, 2008.
2. Programming in C, Reema Theraja, Oxford, 2016, 2nd edition
3. "The C Programming Language", Brian W. Kernighan and Dennis M. Ritchie, Prentice-Hall, 1988
4. Schaum's Outline of Programming with C, Byron S Gottfried, McGraw-Hill Education, 1996

L	T	P	C
3	0	0	3

DATA STRUCTURES

(Common to CSE, IT & allied branches)

Course Outcomes:

At the end of the course, Student will be able to

CO1: Explain the role of linear data structures in organizing and accessing data efficiently in algorithms. (K2)

CO2: Apply linked lists for dynamic data storage, demonstrating understanding of memory allocation (K3)

CO3: Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems. (K3)

CO4: Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between dequeues and priority queues, and apply them appropriately to solve data management challenges. (K3)

CO5: Illustrate scenarios where hashing is advantageous and hash-based solutions for specific problems. (K3)

UNIT I

Introduction to Linear Data Structures: Definition and importance of linear data structures, Abstract data types (ADTs) and their implementation, Overview of time and space complexity analysis for linear data structures. Searching Techniques: Linear & Binary Search, Sorting Techniques: Bubble sort, Selection sort, Insertion Sort

UNIT II

Linked Lists: Singly linked lists: representation and operations, doubly linked lists and circular linked lists, Comparing arrays and linked lists, Applications of linked lists.

UNIT III

Stacks: Introduction to stacks: properties and operations, implementing stacks using arrays and linked lists, Applications of stacks in expression evaluation, backtracking, reversing list etc.

UNIT IV

Queues: Introduction to queues: properties and operations, implementing queues using arrays and linked lists, Applications of queues in breadth-first search, scheduling, etc.

Dequeues: Introduction to dequeues (double-ended queues), Operations on dequeues and their applications.

UNIT V

Trees: Introduction to Trees, Binary Trees, Tree Traversals, Binary Search Tree – Insertion, Deletion & Traversal

Hashing: Brief introduction to hashing and hash functions, Collision resolution techniques: chaining and open addressing, Hash tables: basic implementation and operations, Applications of hashing in unique identifier generation, caching, etc.

Textbooks:

1. Data Structures and algorithm analysis in C, Mark Allen Weiss, Pearson, 2nd Edition.
2. Fundamentals of data structures in C, Ellis Horowitz, Sartaj Sahni, Susan Anderson-Freed, Silicon Press, 2008

Reference Books:

1. Algorithms and Data Structures: The Basic Toolbox by Kurt Mehlhorn and Peter Sanders
2. C Data Structures and Algorithms by Alfred V. Aho, Jeffrey D. Ullman, and John E. Hopcroft
3. Problem Solving with Algorithms and Data Structures" by Brad Miller and David Ranum
4. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein
5. Algorithms in C, Parts 1-5 (Bundle): Fundamentals, Data Structures, Sorting, Searching, and Graph Algorithms" by Robert Sedgewick

L	T	P	C
0	0	2	1

IT WORKSHOP

(Common to All branches of Engineering)

Course Outcomes:

At the end of the course, the student will be able to

CO1: Demonstrate Hardware troubleshooting (K3)

CO2: Illustrate Hardware components and inter dependencies (K3)

CO3: Apply safeguarding techniques to computer systems (K3)

CO4: Prepare Document / Presentation (K3)

CO5: Demonstrate calculations using spreadsheets (K3)

PC Hardware & Software Installation

Task 1: Identify the peripherals of a computer, components in a CPU and its functions. Draw the block diagram of the CPU along with the configuration of each peripheral and submit to your instructor.

Task 2: Disassembling and assembling the PC back to working condition. Lab Instructors should verify the work and follow it up with a Viva. Displaying Video showing the process of assembling a PC. A video would be given as part of the course content.

Task 3: Individual installation of MS windows on the personal computer. Lab instructor should verify the installation and follow it up with a Viva.

Task 4: Linux Installation on the computer. This computer should have windows installed. The system should be configured as dual boot (VMWare) with both Windows and Linux. Lab instructors should verify the installation and follow it up with a Viva

Task 5: Installation of BOSS on the computer. The system should be configured as dual boot (VMWare) with both Windows and BOSS. Lab instructors should verify the installation and follow it up with a Viva

Internet & World Wide Web

Task1: Orientation & Connectivity Boot Camp: Connection to the Local Area Network and accessing the Internet, configuring the TCP/IP setting. Demonstrate to the instructor, how to access the websites and email. If there is no internet connectivity, preparations need to be made by the instructors to simulate the WWW on the LAN.

Task 2: Web Browsers, Surfing the Web: Customization of web browsers with the LAN proxy

settings, bookmarks, search toolbars and pop up blockers. Also, plug-ins like Macromedia Flash and JRE for applets should be configured.

Task 3: Search Engines & Netiquette: Knowledge of search engines and their usage. A few topics to be given to search on Google. This should be demonstrated to the instructors.

Task 4: Cyber Hygiene: Exposure to the various threats on the internet and configuring the computer to be safe on the internet. Browser customization to block pop ups, block active x downloads to avoid viruses and/or worms.

LaTeX and WORD

Task 1 – Word Orientation: The mentor needs to give an overview of LaTeX and Microsoft (MS) office or equivalent (FOSS) tool word: Importance of LaTeX and MS office or equivalent (FOSS) tool Word as word Processors, Details of the four tasks and features that would be covered in each, Using LaTeX and word – Accessing, overview of toolbars, saving files, Using help and resources, rulers, format painter in word.

Task 2: Using LaTeX and Word to create a project certificate. Features to be covered:- Formatting Fonts in word, Drop Cap in word, Applying Text effects, Using Character Spacing, Borders and Colors, Inserting Header and Footer, Using Date and Time option in both LaTeX and Word.

Task 3: Creating project abstract Features to be covered:-Formatting Styles, Inserting table, Bullets and Numbering, Changing Text Direction, Cell alignment, Footnote, Hyperlink, Symbols, Spell Check, Track Changes.

Task 4: Creating a Newsletter: Features to be covered:- Table of Content, Newspaper columns, Images from files and clipart, Drawing toolbar and Word Art, Formatting Images, Textboxes, Paragraphs and Mail Merge in word.

EXCEL

Excel Orientation: The mentor needs to tell the importance of MS office or equivalent (FOSS) tool Excel as a Spreadsheet tool, give the details of the four tasks and features that would be covered in each. Using Excel – Accessing, overview of toolbars, saving excel files, Using help and resources.

Task 1: Creating a Scheduler - Features to be covered: Gridlines, Format Cells, Summation, auto fill, Formatting Text

Task 2: Calculating GPA - Features to be covered:- Cell Referencing, Formulae in excel – average, std. deviation, Charts, Renaming and Inserting worksheets, Hyper linking, Count function

LOOKUP/VLOOKUP

Task 3: Split cells, freeze panes, group and outline, Sorting, Boolean and logical operators, Conditional formatting

POWER POINT

Task 1: Working on basic power point utilities and tools to create basic power point presentations. PPT Orientation, Slide Layouts, Inserting Text, Word Art, Formatting Text, Bullets and Numbering, Auto Shapes, Lines and Arrows in PowerPoint.

Task 2: Interactive presentations - Hyperlinks, Inserting –Images, Clip Art, Audio, Video, Objects, Tables and Charts.

Task 3: Master Layouts (slide, template, and notes), Types of views (basic, presentation, slide slotter, notes etc), and Inserting – Background, textures, Design Templates, Hidden slides.

AI TOOLS – ChatGPT

Task 1: Prompt Engineering: Experiment with different types of prompts to see how the model responds. Try asking questions, starting conversations, or even providing incomplete sentences to see how the model completes them.

- Ex: Prompt: "You are a knowledgeable AI. Please answer the following question: What is the capital of France?"

Task 2: Creative Writing: Use the model as a writing assistant. Provide the beginning of a story or a description of a scene, and let the model generate the rest of the content. This can be a fun way to brainstorm creative ideas

- Ex: Prompt: "In a world where gravity suddenly stopped working, people started floating upwards. Write a story about how society adapted to this new reality."

Task 3: Language Translation: Experiment with translation tasks by providing a sentence in one language and asking the model to translate it into another language. Compare the output to see how accurate and fluent the translations are.

- Ex: Prompt: "Translate the following English sentence to French: 'Hello, how are you doing today?'"

Reference Books:

1. Comdex Information Technology course tool kit, Vikas Gupta, WILEY Dream tech, 2003
2. The Complete Computer upgrade and repair book, Cheryl A Schmidt, WILEY Dream tech, 2013, 3rd edition
3. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education, 2012, 2nd edition

4. PC Hardware - A Handbook, Kate J. Chase, PHI (Microsoft)
5. LaTeX Companion, Leslie Lamport, PHI/Pearson.
6. IT Essentials PC Hardware and Software Companion Guide, David Anfinson and Ken Quamme.
– CISCO Press, Pearson Education, 3rd edition
7. IT Essentials PC Hardware and Software Labs and Study Guide, Patrick Regan– CISCO Press,
Pearson Education, 3rd edition
8. How Computers Work, Ron White, Tim Edward Downs, Que Publications

L	T	P	C
0	0	3	1.5

DATA STRUCTURES LAB

(Common to CSE, IT & allied branches)

Course Outcomes:

At the end of the course, Student will be able to

CO1: Explain the role of linear data structures in organizing and accessing data efficiently in algorithms. (K2)

CO2: Apply linked lists for dynamic data storage, demonstrating understanding of memory allocation (K3)

CO3: Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems. (K3)

CO4: Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between dequeues and priority queues, and apply them appropriately to solve data management challenges. (K3)

CO5: Illustrate scenarios where hashing is advantageous and hash-based solutions for specific problems. (K3)

List of Experiments:

Exercise 1: Array Manipulation

- Write a program to reverse an array.
- C Programs to implement the Searching Techniques – Linear & Binary Search
- C Programs to implement Sorting Techniques – Bubble, Selection and Insertion Sort

Exercise 2: Linked List Implementation

- Implement a singly linked list and perform insertion and deletion operations.
- Develop a program to reverse a linked list iteratively and recursively.
- Solve problems involving linked list traversal and manipulation.

Exercise 3: Linked List Applications

- Create a program to detect and remove duplicates from a linked list.
- Implement a linked list to represent polynomials and perform addition.
- Implement a double-ended queue (deque) with essential operations.

Exercise 4: Double Linked List Implementation

- i) Implement a doubly linked list and perform various operations to understand its properties and applications.
- ii) Implement a circular linked list and perform insertion, deletion, and traversal.

Exercise 5: Stack Operations

- i) Implement a stack using arrays and linked lists.
- ii) Write a program to evaluate a postfix expression using a stack.
- iii) Implement a program to check for balanced parentheses using a stack.

Exercise 6: Queue Operations

- i) Implement a queue using arrays and linked lists.
- ii) Develop a program to simulate a simple printer queue system.
- iii) Solve problems involving circular queues.

Exercise 7: Stack and Queue Applications

- i) Use a stack to evaluate an infix expression and convert it to postfix.
- ii) Create a program to determine whether a given string is a palindrome or not.
- iii) Implement a stack or queue to perform comparison and check for symmetry.

Exercise 8: Binary Search Tree

- i) Implementing a BST using Linked List.
- ii) Traversing of BST.

Exercise 9: Hashing

- i) Implement a hash table with collision resolution techniques.
- ii) Write a program to implement a simple cache using hashing.

Textbooks:

1. Data Structures and algorithm analysis in C, Mark Allen Weiss, Pearson, 2nd Edition.
2. Fundamentals of data structures in C, Ellis Horowitz, Sartaj Sahni, Susan Anderson-Freed, Silicon Press, 2008

Reference Books:

1. Algorithms and Data Structures: The Basic Toolbox by Kurt Mehlhorn and Peter Sanders
2. C Data Structures and Algorithms by Alfred V. Aho, Jeffrey D. Ullman, and John E. Hopcroft

3. Problem Solving with Algorithms and Data Structures" by Brad Miller and David Ranum
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